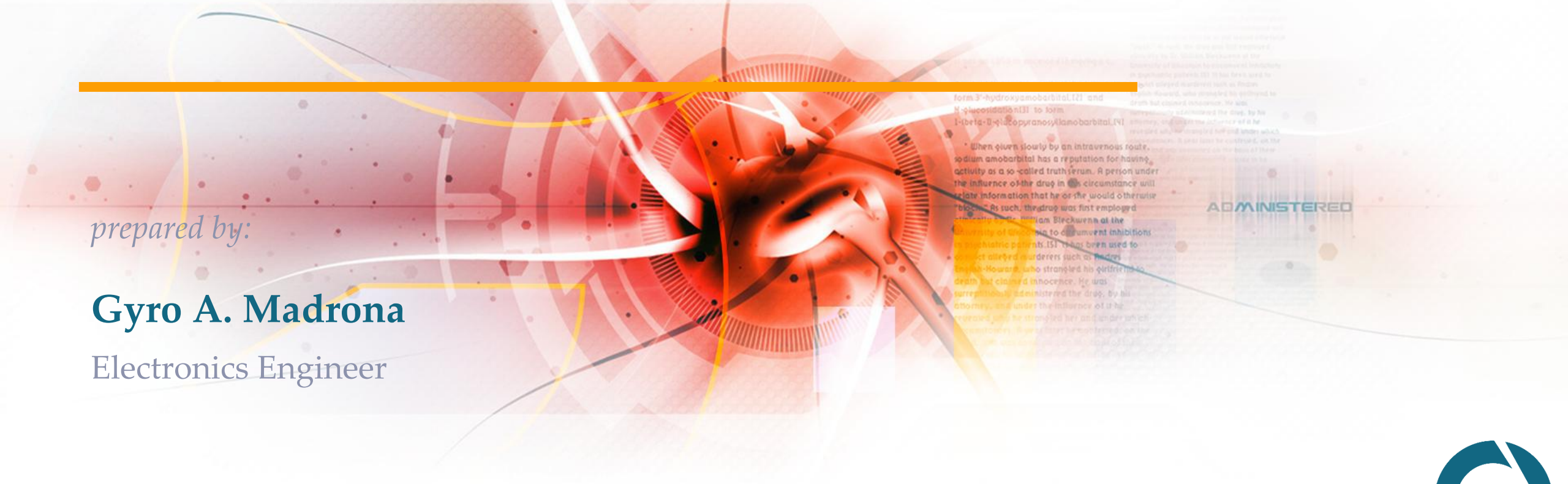
THE SPHERE

SOLID MENSURATION



prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

0

ABSENT

3

45-min LATE

5 max

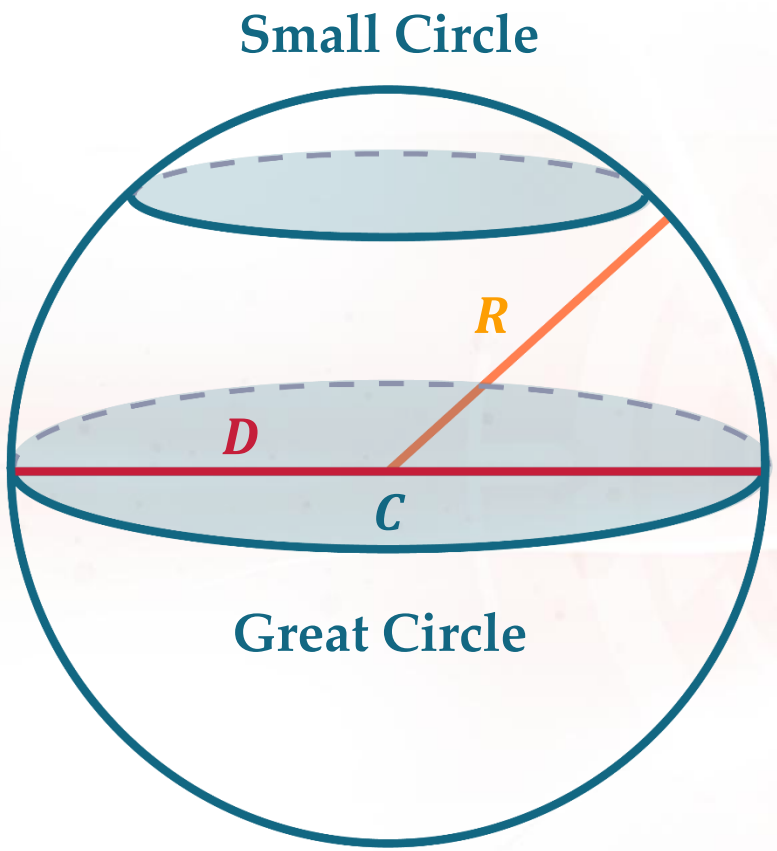
15-min LATE

TOPIC OUTLINE

Properties of a Sphere

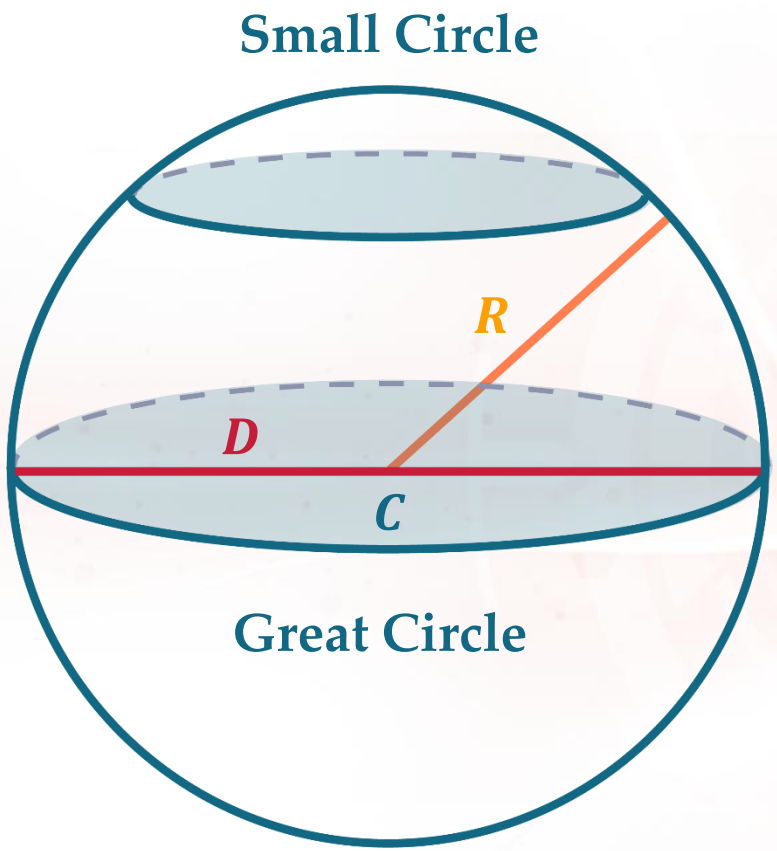
- Surface Area
- Volume

PROPERTIES OF A SPHERE



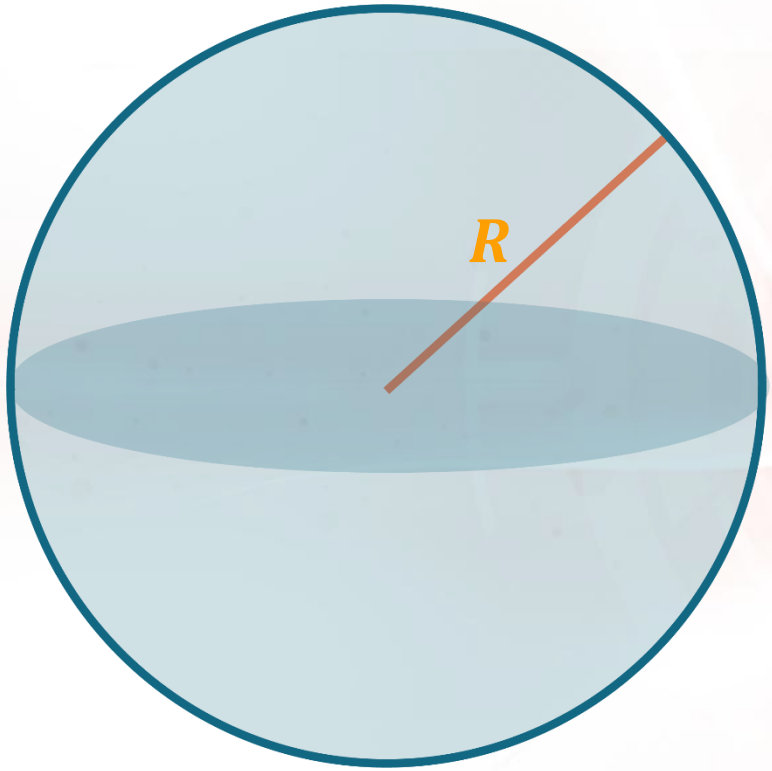
A **sphere** is a solid bounded by a closed surface every point of which is **equidistant, R** from a fixed point called the **center, C**.

PROPERTIES



1. Every plane section of a sphere is a circle. If the plane contains a diameter of the sphere, the section is a **great circle**; otherwise, the section is a **small circle**.
2. The **axis** of a sphere is the **diameter, *D*** of the sphere perpendicular to the plane of the circle.
3. The **radius, *R*** of a great circle is equal to the radius of the sphere.

AREA OF A SPHERE



The Surface Area **S**

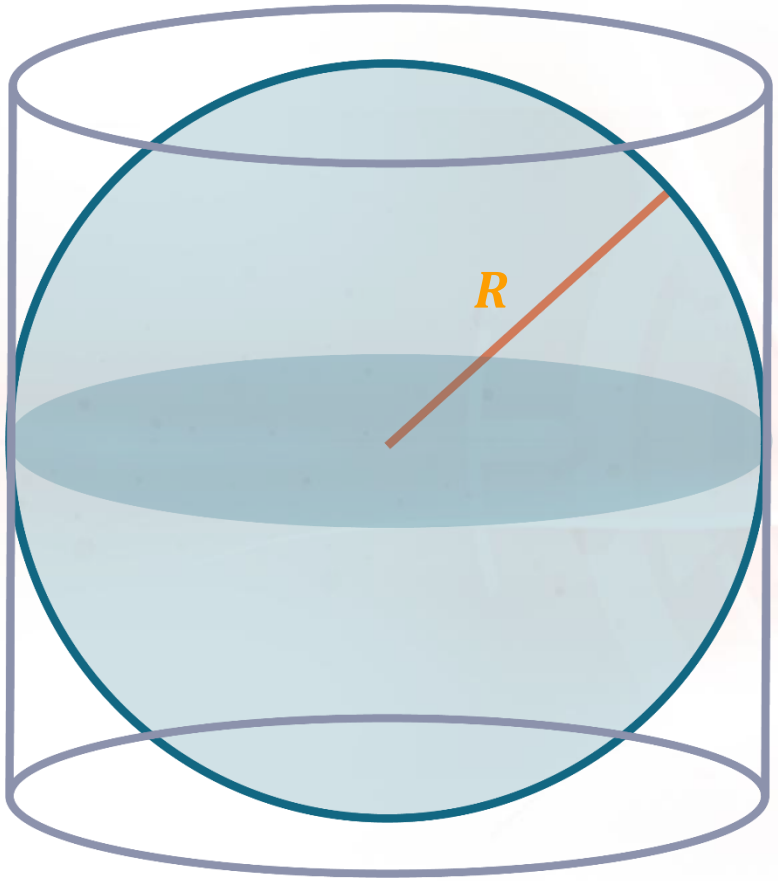
$$S = area\ of\ 4\ great\ circles$$

$$\boxed{S = 4\pi r^2}$$

where

r = radius of the sphere

Consider a **solid sphere** with a **radius** of **5** cm. Calculate the **cost** of painting it, assuming that the cost of paint is **$0.08 per square centimeter**.



The Volume **V**

$$V = \frac{4}{3}Bh$$

$$\boxed{V = \frac{4}{3}\pi r^3}$$

where

r = radius of the sphere

Find the **weight** of a **snowball 4** ft. in **diameter** if the wet compact snow of which this ball is made weighs **30 lb. per cu. ft**.

An **iron ball 4** in. in **diameter** weighs **9 lb**. Find the **weight** of an **iron shell 2 in. thick** whose **external** diameter is **20 in.**

SEATWORK